

IBM PC110 U6 Firmware Description

M38223E4HP power, LCD, serial, and resistive signature-pad controller

Binary: M38223E4HP@QFP80.BIN | SHA-256: 96c6e37cfa52f30b... | ROM map: \$C080-\$FFFD

Executive summary

U6 is best understood as the PC110 supervisory microcontroller. The ROM banner identifies it as "M3822X POWER SENSE MICON FIRMWARE Rev 8". Static disassembly plus the schematic and prior HTML annotations show that it also drives the resistive signature pad: P10-P13 switch VCC/GND onto the touch X/Y rails, while P65/P66 are sampled through the ADC. The firmware samples both axes, filters and normalizes the coordinates, then supplies a three-byte packet to the Bowman/ASIC side over P0 with a P1.5 strobe.

1. Source material and confidence level

Source	Role	Notes
M38223E4HP@QFP80.BIN	Authoritative firmware image used for all byte-level decoding.	16254 bytes, SHA-256 96c6e37cfa52f30b303db70c2036cbf21e6e1bb638c5eb11343ab161db3c9cc0
PC110-Singlesheet-Schematic.pdf	Board-level correlation for U6 pins, LCD COM/SEG pins, PWRGD/RESET, and P10-P13/P65/P66 placement.	Used for hardware interpretation.
M38223E4HP@QFP80.BIN.html	Unfinished prior reverse-engineering pass. Useful labels and touch annotations, but not byte-identical to the uploaded BIN.	Used as advisory only.
Generated clean ASM	Fresh cleaned listing derived from the BIN and updated with high-confidence comments.	U6_M38223E4HP_clean_disassembly.asm

Confidence note. The touch-pad path is high confidence because pin mapping, ADC channel selection, and code behavior all agree. Exact names for many general power-state variables remain provisional because this is still static analysis, not a live hardware trace.

2. ROM layout and vector table

ROM mapping. File offset 0 maps to ROM address \$C080; the final byte maps to \$FFFD. The reset vector at \$FFFC/\$FFFD points to \$E6B1. The ASCII banner starts at \$C080 and the first executable stub begins at \$C0C6.

Vector	Word addr	Target	Current interpretation
BRK	\$FFDC	\$E8FC	unused/default
ADT_ADC	\$FFDE	\$D612	ADC completion interrupt
KEY_WAKEUP	\$FFE0	\$E8FC	unused/default
INT3	\$FFE2	\$D190	external interrupt 3
INT2	\$FFE4	\$D177	external interrupt 2
TIMER1	\$FFE6	\$E8FC	unused/default
CNTR1	\$FFE8	\$E8FC	unused/default
CNTR0	\$FFEA	\$E8FC	unused/default
TIMER3	\$FFEC	\$E307	P4.0 sampling/history path
TIMER2	\$FFEE	\$E366	periodic service
TIMER_Y	\$FFF0	\$D8D4	main periodic scheduler
TIMER_X	\$FFF2	\$E8FC	unused/default
SERIAL_TX	\$FFF4	\$E2AA	serial transmit
SERIAL_RX	\$FFF6	\$E2CF	serial receive
INT1	\$FFF8	\$DC61	Bowman/ASIC touch byte request
INT0	\$FFFA	\$D1AF	external power/LCD related path
RESET	\$FFFC	\$E6B1	reset entry

3. U6 hardware role from schematic

Schematic correlation. U6 is labeled M38223E4HP and exposes RESET, PWRGD, Xin/Xout, VREF, VL1/VL2/VL3, COM0-COM3, SEG pins, and ports P0-P7. That matches an M3822 LCD microcontroller used as a low-power supervisor rather than a generic glue part.

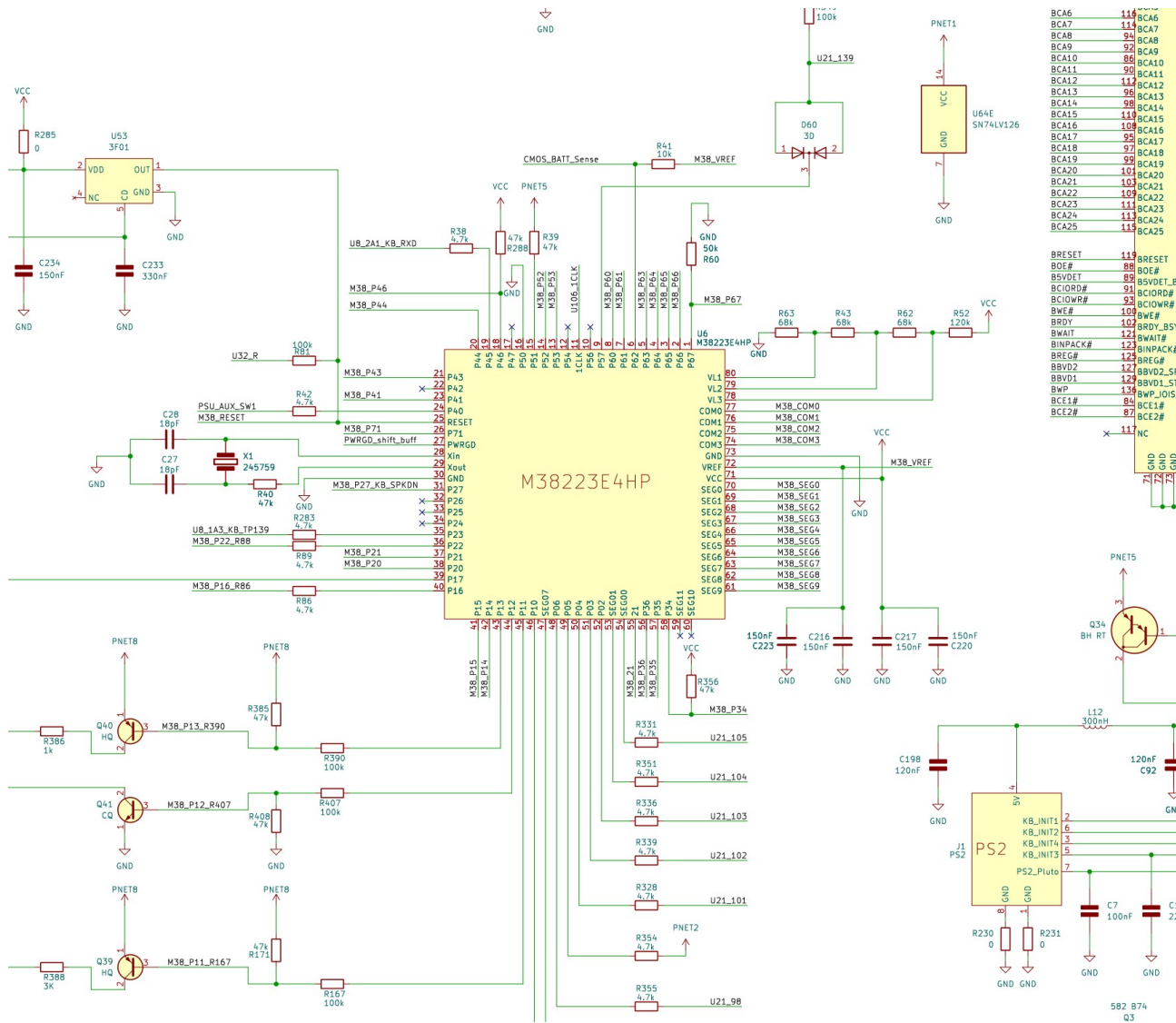


Figure 1. U6 schematic crop showing the M38223E4HP pins, including P10-P13, P65/P66, COM/SEG, VREF, RESET, and PWRGD.

3.1 Pin/function summary

U6 pins/signals	Hardware meaning	Firmware evidence
P0.0-P0.7	Parallel byte path to Bowman/ASIC	Used by \$DC6F to place packet bytes on P0.
P1.0/P10	Touch_Y- to GND switch	Set high during Y-drive phase.
P1.1/P11	Touch_Y+ to VCC/PWR switch	Cleared low during Y-drive phase; active-low power switch.
P1.2/P12	Touch_X- to GND switch	Set high during X-drive phase.
P1.3/P13	Touch_X+ to VCC/PWR switch	Cleared low during X-drive phase; active-low power switch.
P1.5	Bowman/ASIC byte strobe	Pulsed low then high after writing P0.
P6.5/AN5	Touch analog sample, phase 2	Selected by LDA #\$05 then ADC start.
P6.6/AN6	Touch analog sample, phase 1	Selected by LDA #\$06 then ADC start.
P7.0/PWRGD	Power-good related signal	Used in power/LCD state logic.
COM/SEG/VL/REF	LCD drive and bias pins	Firmware writes LCD RAM and LM/SEG registers.

4. Reset and main loop

Reset entry at \$E6B1. The reset handler disables interrupts, clears decimal/test/overflow state, jumps into the SFR initialization loop, clears low RAM and page-1 RAM, sets the stack pointer to \$BE, then enters the state-machine path.

```
reset_entry:
E6B1: sei
E6B2: cld
E6B3: clt
E6B4: clv
E6B5: jmp reset_init_start

reset_sfr_init_loop:
E6CE: ldy sfr_init_table,X      ; register address
E6D1: cpy #$40                 ; $40 sentinel ends the table
E6D6: lda sfr_init_table,X      ; register value
E6D9: sta $0000,Y              ; write SFR
```

Main service loop. After the banner, the firmware enters a loop that waits on ADC completion, services power/mode state, maintains LCD state, and lets the interrupt handlers perform time-critical work. The loop delegates to a state dispatcher at \$CD17.

5. Resistive signature-pad acquisition

Hardware strategy. The MCU alternately drives one resistive axis as a voltage divider and samples the opposite axis through the ADC. Each axis change is followed by eight discarded ADC conversions to let analog switches, op-amps, and panel capacitance settle. The ninth conversion is treated as the final coordinate sample.

Step	Action	Port operation	ADC/data result
1	Drive X rail	P1.2 high = X- to GND; P1.3 low = X+ to VCC/PWR	ADC channel 6 / AN6 / P66
2	Settle	\$DA = 8; ADC ISR discards 8 samples	Restart same channel after each discard
3	Latch first axis	Disconnect X drive; store AD to \$D0	\$D0 = raw first-axis value
4	Drive Y rail	P1.0 high = Y- to GND; P1.1 low = Y+ to VCC/PWR	ADC channel 5 / AN5 / P65
5	Settle	\$DA = 8; ADC ISR discards 8 samples	Restart same channel after each discard
6	Latch and normalize	Disconnect Y drive; threshold both axes	\$CE = axis0, \$CF = inverted axis1

```

touch_setup_x_phase_drive_x_read_an6:
D849: ldm #$08,$DA<TouchSettleCountdown>
D84E: seb 2,$02<P1>          ; P12/P1.2: Touch_X- -> GND
D850: clb 3,$02<P1>          ; P13/P1.3: Touch_X+ -> VCC/PWR
D858: seb 5,$81<ADC_Global_State>
D85C: lda #$06              ; ADC channel AN6/P66
D85E: jsr adc_start_channel_vref

```

```

touch_setup_y_phase_drive_y_read_an5:
D862: ldm #$08,$DA<TouchSettleCountdown>
D865: seb 0,$02<P1>          ; P10/P1.0: Touch_Y- -> GND
D867: clb 1,$02<P1>          ; P11/P1.1: Touch_Y+ -> VCC/PWR
D86D: seb 6,$81<ADC_Global_State>
D871: lda #$05              ; ADC channel AN5/P65
D873: jsr adc_start_channel_vref

```

6. Touch coordinate filtering and packet format

Coordinate variables. \$CE holds the first-axis normalized coordinate, likely X. \$CF holds the second-axis normalized coordinate, likely Y, and is inverted with COM so that both reported axes follow the same physical direction convention.

RAM byte	Role	Evidence/interpretation
\$CE	First-axis normalized coordinate	Raw AN6/P66 is thresholded; invalid values become 0.
\$CF	Second-axis normalized coordinate, inverted	Raw AN5/P65 is thresholded; invalid values become \$FF before inversion.
\$D1/\$D2	Previous accepted coordinate baseline	Used by the delta filter at \$DC0D.
\$D5	Packet header/status	Contains valid bits plus high bits from X/Y.
\$D6	X low 7 bits	Produced by shifting X left/right around the carry.
\$D7	Y low 7 bits	Produced similarly for Y.
\$D8	Transmit countdown	Used by \$DC6F, but see DBFE anomaly below.
\$D9	Transmit byte index	Selects D5/D6/D7 byte for output.

```

touch_pack_coordinate_packet:
DBD6: ldm #$00,$D5<TouchPktHeader>
DBD9: asl A                ; carry = X bit7
DBDA: rol $D5              ; header receives X high bit
DBDD: sta $D6              ; X low 7 bits
DBDF: lda $CF              ; Y
DBE1: asl A                ; carry = Y bit7
DBE2: rol $D5              ; header receives Y high bit
DBE5: sta $D7              ; Y low 7 bits
DBED: seb 7,$D5            ; header/valid bit
DBEF: seb 0,$D5            ; header/valid bit
DC6F: lda $D8              ; bytes pending?
DC77: lda $D5,X            ; send D5, then D6, then D7
DC79: sta $00<P0>          ; output data byte
DC7B: clb 5,$02<P1>
DC7D: seb 5,$02<P1>        ; strobe

```

DBFE anomaly to verify

The uploaded BIN contains bytes 3C 03 D0 at \$DBFE, decoded as LDM #\$03,\$D0. The older HTML attempt labels the same logical point as writing \$D8, which would fit the transmit countdown used at \$DC6F. Because the BIN is the source of truth, the clean ASM records the uploaded bytes but flags this as a hardware-probing target.

7. Other firmware subsystems

Subsystem	Firmware/hardware clues	Interpretation
-----------	-------------------------	----------------

LCD drive	\$40-\$4F RAM, LM at \$39, SEG at \$38, COM/SEG pins	Display/icon updates and enable/disable helpers exist around \$D1BB-\$D1C9 and display routines around \$D1CD onward.
Power and reset supervision	PWRGD, P7, P2/P4/P5 paths, reset fallback	The firmware banner says power-sense; the scheduler contains conditions that jump back to reset_entry.
ADC power/battery sensing	ADCON/AD plus P6 analog channels	ADC ISR handles several states beyond touch: battery, dock, charge/current, CMOS battery style channels.
Serial I/O	TB/RB \$18, SIOSTS/SIOCON, TX/RX vectors	TX ISR at \$E2AA, RX ISR at \$E2CF; RX stores data into RAM buffer at page 1.
Timers	Timer Y, Timer 2, Timer 3	Timer Y acts as the broad scheduler. Timer 3 samples P4.0. Timer 2 participates in periodic service paths.

8. Important routine map

Address	Clean label	Meaning
\$C0C6	post_banner_startup	Executable stub immediately after the banner.
\$C0D8	main_service_loop	Main polling/service loop.
\$CD17	state_dispatch	Power/mode state-machine dispatcher.
\$D612	isr_adc_complete	ADC interrupt entry and ADC state dispatch.
\$D7CD	adc_start_channel_vref	Starts ADC conversion with VREF enabled.
\$D849	touch_setup_x_phase_drive_x_read_an6	Drive X rail and read AN6/P66 after settling.
\$D862	touch_setup_y_phase_drive_y_read_an5	Drive Y rail and read AN5/P65 after settling.
\$D877	touch_x_adc_tick	Discard settling samples then store raw first-axis sample.
\$D897	touch_y_adc_tick	Discard settling samples then produce \$CE/\$CF coordinates.
\$DB94	touch_gate_validate_and_pack	Gates touch packet generation.
\$DBD6	touch_pack_coordinate_packet	Builds three-byte packet in \$D5-\$D7.
\$DC61	isr_bowman_touch_request	INT1 handler for Bowman/ASIC byte requests.
\$DC6F	bowman_touch_tx_next_byte	Writes P0 byte and pulses P1.5.
\$D8D4	isr_timer_y_scheduler	Periodic scheduler and touch packet production path.
\$E2AA/\$E2CF	serial TX/RX ISRs	Serial communication handlers.
\$E307	isr_timer3_p4_sampler	Timer 3 input sampler/history path.
\$E6B1	reset_entry	Reset vector target and full MCU initialization.

9. Practical hardware validation plan

Question	Suggested check
Confirm DBFE countdown target	Probe INT1/P0/P1.5 during a touch. If packets never transmit unless \$D8 is set elsewhere, compare against another ROM dump or patch DBFE in a test fixture.
Confirm axis orientation	Log P65/P66 raw values while pressing known corners; verify whether \$CE truly maps to X and \$CF to Y after inversion.
Measure settle timing	Scope P1.0-P1.3 and ADC input nodes during a touch; verify the eight discarded conversion cadence.
Packet decode	Capture P0 and P1.5. Decode D5/D6/D7 as header plus 7-bit X/Y payload and compare with physical motion.
Power/LCD interactions	Observe PWRGD, RESET, COM/SEG, and touch enable while changing main power states.

10. Known limitations

Limitation	Meaning
Static analysis only	The clean ASM is a static listing, not an emulator trace. Some computed jumps, RAM state bits, and table boundaries still need live observation.
HTML attempt was advisory	The older HTML annotations helped with names, but the uploaded BIN is treated as authoritative wherever bytes disagree.
Axis naming is inferred	\$CE is called the first axis / likely X and \$CF the second axis / likely Y. Physical corner tests should confirm exact orientation.
Untraced regions	The listing keeps linear disassembly for coverage, but untraced areas may be data tables decoded as instructions.
Best validation target	Capture P1.0-P1.3, P6.5/P6.6, INT1, P0, and P1.5 while drawing on the pad; this should validate the entire acquisition-to-packet path.

11. Deliverables

Clean ASM. U6_M38223E4HP_clean_disassembly.asm is the cleaned full listing. It keeps every decoded ROM byte visible while reducing generic noise and adding high-confidence comments for the touch and supervisory paths.

This PDF. The PDF is intended as a readable engineering description, not a formal proof. It should be read together with the clean ASM and, for final validation, hardware traces from P1.0-P1.3, P6.5/P6.6, P0, P1.5, and INT1.